

- 27** In the photoelectric effect, light falling on a metal surface causes electrons to be ejected from the surface.

Which statement is correct?

- A** Electrons are ejected only if the wavelength of the incident light is greater than some minimum value.
- B** The maximum energy of the electrons is independent of the intensity of the incident light.
- C** The maximum energy of the electrons is independent of the type of metal.
- D** The waves associated with the ejected electrons have the same wavelength as the incident light.

27	B	A: maximum wavelength / minimum frequency rather than minimum wavelength C: $hf = E_{k\max} + \phi$ and ϕ depends on type of metal D: the electrons do not have the same energy as the incident photons, even then you need to use De Broglie's equation.
-----------	----------	--

[Turn over

29 Electrons with velocity v travel through a vacuum and are incident on a thin carbon film, as